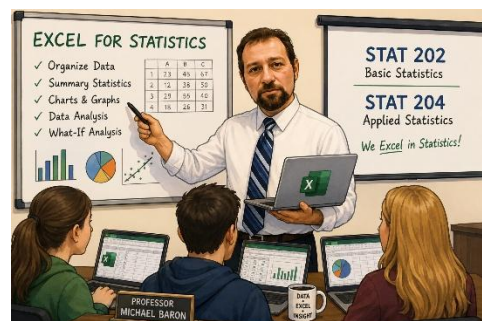


Analysis of Variance (ANOVA)

Given the raw data, we can use Excel's Data Analysis Toolpak for both the one-way and the two-way ANOVA.



One-Way ANOVA on Data Analysis Toolpak

For One-Way ANOVA, use the “ANOVA: Single Factor” tool. The groups must be organized in columns or rows.

A screenshot of the Microsoft Excel interface. The 'DATA' tab is selected in the ribbon. The 'Data Analysis' group is expanded, showing various tools. 'Anova: Single Factor' is highlighted with a yellow oval. The background shows a spreadsheet with data for different meats: Kidney, Shrimp, Chicken Liver, Salmon, and Beef, with numerical values in columns A through E. Rows 1 through 11 are visible. The spreadsheet title is 'CatFood.xlsx - Excel'.

The screenshot shows the 'Anova: Single Factor' dialog box in Microsoft Excel. The dialog box is configured with the following settings:

- Input Range:** \$A\$1:\$E\$11
- Grouped By:** Columns (selected)
- Labels in First Row:** ☒
- Alpha:** 0.05
- Output options:**
 - ☐ Output Range:
 - ☒ New Worksheet Ply:
 - ☐ New Workbook

The background data table is as follows:

	A	B	C	D	E
1	Kidney	Shrimp	Chicken Liver	Salmon	Beef
2	2.37	2.26	2.29	1.79	2.09
3	2.62	2.69	2.23	2.33	1.87
4	2.31	2.25	2.41	1.96	1.67
5	2.47	2.45	2.68	2.05	1.64
6	2.59	2.34	2.25	2.26	2.16
7	2.62	2.37	2.17	2.24	1.75
8	2.34	2.22	2.37	1.96	1.18
9	2.47	2.56	2.26	1.58	1.92
10	2.45	2.36	2.45	2.18	1.32
11	2.32	2.59	2.57	1.93	1.94

The resulting ANOVA output contains descriptive statistics (group sample means and sample variances) and the standard ANOVA table with sums of squares, degrees of freedom, mean squares, and the F-statistic with the associated p-value and rejection boundary.

FILE

HOME

INSERT

PAGE LAYOUT

FORMULAS

DATA

REVIEW

VIEW

From Access

From Web

From Text

From Other Sources

Existing Connections

Refresh All

Connections

Properties

Edit Links

A Z

Z A

Sort

Filter

Clear

Reap

Adva

Get External Data

Connections

Sort & Filter

	A	B	C	D	E	F	G	H
1	Anova: Single Factor							
2								
3	SUMMARY							
4	Groups	Count	Sum	Average	Variance			
5	Kidney	10	24.56	2.456	0.01476			
6	Shrimp	10	24.09	2.409	0.025343			
7	Chicken Liver	10	23.68	2.368	0.026284			
8	Salmon	10	20.28	2.028	0.054418			
9	Beef	10	17.54	1.754	0.099027			
10								
11								
12	ANOVA							
13	Source of Variation	SS	df	MS	F	P-value	F crit	
14	Between Groups	3.65896	4	0.91474	20.80541	9.15E-10	2.578739	
15	Within Groups	1.97849	45	0.043966				
16								
17	Total	5.63745	49					
18								
19								

Two-way ANOVA on Data Analysis Toolpak

For the two-way ANOVA, there are two options – with and without replications. We'll mostly have at least two replications n' for each combination of factor levels, and therefore, we'll use "ANOVA: two-factor with replications".

Excel ribbon: FILE, HOME, INSERT, PAGE LAYOUT, FORMULAS, DATA, REVIEW, VIEW, PAMF

DATA ribbon options: Get External Data (From Access, From Web, From Text, From Other Sources), Connections (Connections, Properties, Edit Links, Refresh All), Sort & Filter (Sort, Filter, Clear, Reapply, Advanced)

	A	B	C	D	E	F	G	H	I	J
1	Type	Early	Late							
2	Continental	1.2	-2.5							
3	Continental	2.1	3.0							
4	Continental	3.3	-0.2							
5	Continental	4.4	1.2							
6	Continental	3.4	1.2							
7	Continental	5.3	0.7							
8	Continental	2.2	-1.3							
9	Continental	1.0	0.2							
10	Continental	5.4	-0.5							
11	Continental	1.4	3.8							
12	American	4.4	6.0							
13	American	1.1	2.3							
14	American	4.8	4.2							

Data Analysis dialog box options:

- Analysis Tools
 - Anova: Single Factor
 - Anova: Two-Factor With Replication**
 - Anova: Two-Factor Without Replication
 - Correlation
 - Covariance
 - Descriptive Statistics
 - Exponential Smoothing
 - F-Test Two-Sample for Variances
 - Fourier Analysis
 - Histogram

Buttons: OK, Cancel, Help

Excel will only handle *balanced* two-way ANOVA, where the number of replications n' is the same for each combination. This number, n' , has to be entered as “Rows per sample”:

Excel ribbon: FILE, HOME, INSERT, PAGE LAYOUT, FORMULAS, DATA, REVIEW, VIEW, PAMF/

DATA ribbon options: From Access, From Web, From Text, From Other Sources, Existing Connections, Refresh All, Connections, Properties, Edit Links, Sort, Filter, Clear, Reapply, Advanced.

	A	B	C	D	E	F	G	H	I	J
1	Type	Early	Late							
2	Continental	1.2	-2.5							
3	Continental	2.1	3.0							
4	Continental	3.3	-0.2							
5	Continental	4.4	1.2							
6	Continental	3.4	1.2							
7	Continental	5.3	0.7							
8	Continental	2.2	-1.3							
9	Continental	1.0	0.2							
10	Continental	5.4	-0.5							
11	Continental	1.4	3.8							
12	American	4.4	6.0							
13	American	1.1	2.3							
14	American	4.8	4.2							
15	American	7.1	3.8							
16	American	6.7	5.5							
17	American	5.6	1.8							
18	American	9.5	5.1							
19	American	4.1	4.2							
20	American	7.9	4.9							
21	American	9.4	4.0							
22										
23										

Anova: Two-Factor With Replication

Input
 Input Range: SAS1:SCS21
 Rows per sample: 10
 Alpha: 0.05

Output options
☐ Output Range:
☒ New Worksheet Ply:
☐ New Workbook

OK, Cancel, Help

The output contains descriptive statistics for each combination of factor levels, sums of squares, degrees of freedom, and mean squares for each main factor and the interaction, the corresponding F-statistics, and their P-values.

In the current example, one factor is the time of the **flight** – early or late, and the other factor is the **airline** – American or Continental. Each factor has two levels, $r=2$ and $c=2$, and each

combination of levels has $n'=10$ replications, for a total sample size of $n=40$.

	A	B	C	D	E	F	G	H
1	Anova: Two-Factor With Replication							
2								
3	SUMMARY	Early	Late	Total				
4	Continental							
5	Count	10	10	20				
6	Sum	29.7	5.6	35.3				
7	Average	2.97	0.56	1.765				
8	Variance	2.72233	3.54933	4.49924				
9								
10	American							
11	Count	10	10	20				
12	Sum	60.6	41.8	102.4				
13	Average	6.06	4.18	5.12				
14	Variance	6.76267	1.75511	4.96484				
15								
16	Total							
17	Count	20	20					
18	Sum	90.3	47.4					
19	Average	4.515	2.37					
20	Variance	7.00555	5.96116					
21								
22								
23	ANOVA							
24	Source of Variation	SS	df	MS	F	P-value	F crit	
25	Sample	112.56	1	112.56	30.4434	3.1E-06	4.11317	
26	Columns	46.0103	1	46.0103	12.4441	0.00117	4.11317	
27	Interaction	0.70225	1	0.70225	0.18993	0.66557	4.11317	
28	Within	133.105	36	3.69736				
29	Total	292.378	39					
30								

Here, the row factor (airline) is listed as “Sample”, and it is found significant with a very low p-value of 0.0000031. The column factor “Time of the flight” is listed as “Columns”, and it is found significant too, with a p-value of 0.00117. The interaction effect is not significant, with a p-value of 0.66557. Each F-statistic is obtained as a ratio of the corresponding MS and the Within MS which is also known as the Mean Squared Error (MSE).

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		Kidney	Shrimp	Liver	Salmon	Beef							
2		2.37	2.26	2.29	1.79	2.09							
3		2.62	2.69	2.23	2.33	1.87							
4		2.31	2.25	2.41	1.96	1.67							
5		2.47	2.45	2.68	2.05	1.64							
6		2.59	2.34	2.25	2.26	2.16							
7		2.62	2.37	2.17	2.24	1.75							
8		2.34	2.22	2.37	1.96	1.18							
9		2.47	2.56	2.26	1.58	1.92							
10		2.45	2.36	2.45	2.18	1.32							
11		2.32	2.59	2.57	1.93	1.94							
12								Total					
13	=AVERAGE(B2:B11)	2.456	2.409	2.368	2.028	1.754	=AVERAGE(B2:F11)	2.203	Grand mean				
14	=VAR(B2:B11)	0.0148	0.0253	0.0263	0.0544	0.099							
15	=COUNT(B2:B11)	10	10	10	10	10							
16													
17	=B15*(B13-\$H13)^2	0.6401	0.4244	0.2722	0.3063	2.016	=SUM(B17:F17)	3.65896	SSA	DF	MS	Fstat	P-value
18	=(B15-1)*B14	0.1328	0.2281	0.2366	0.4898	0.8912	=SUM(B18:F18)	1.97849	SSW	4	0.9147	20.80541	9.149E-10
19										45	0.044		

These SSA and SSW are computed in H17-18 above. Divide them by their corresponding degrees of freedom to get MSA and MSW, then compute $F_{\text{stat}} = \text{MSA}/\text{MSW}$, and finally, the P-value by = **FDIST(L17,J17,J18)**.

Two-way ANOVA manually

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	A
1	Type	Early	Late		mu(I,j)			mu(I,.)			mu(.,j)			mu(.,.)			SSA			SSB			SSAB			SSE		
2	Continental	1.2	-2.5		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		3.13	9.364	
3	Continental	2.1	3.0		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.76	5.954	
4	Continental	3.3	-0.2		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.11	0.578	
5	Continental	4.4	1.2		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		2.04	0.41	
6	Continental	3.4	1.2		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.18	0.41	
7	Continental	5.3	0.7		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		5.43	0.02	
8	Continental	2.2	-1.3		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.59	3.46	
9	Continental	1.0	0.2		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		3.88	0.13	
10	Continental	5.4	-0.5		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		5.9	1.124	
11	Continental	1.4	3.8		3.0	0.6		1.8	1.8		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		2.46	10.5	
12	American	4.4	6.0		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		2.76	3.312	
13	American	1.1	2.3		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		24.6	3.534	
14	American	4.8	4.2		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		1.59	4E-04	
15	American	7.1	3.8		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		1.08	0.144	
16	American	6.7	5.5		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.41	1.742	
17	American	5.6	1.8		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		0.21	5.664	
18	American	9.5	5.1		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		11.8	0.846	
19	American	4.1	4.2		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		3.84	4E-04	
20	American	7.9	4.9		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		3.39	0.518	
21	American	9.4	4.0		6.1	4.2		5.1	5.1		4.5	2.4		3.4	3.4		2.81	2.814		1.15	1.1503		0.02	0.018		11.2	0.032	
22																												
23																	SSA=	112.56		SSB=	46.01		SSAB	0.702		SSE=	133.1	
24																	dfA=	1		dfB=	1		dfAB	1		dfE=	36	
25																	MSA=	112.56		MSB=	46.01		MSA	0.702		MSE	3.697	
26																												
27																	Fstat	30.443			12.444			0.19				
28																	Pvalue	3E-06			0.0012			0.666				

Here,

- The raw data are in columns A-C.
- Sample means $\bar{X}_{i.}$, $\bar{X}_{.j}$, and $\bar{X}_{i,j}$ are computed in columns E-F, H-I, and K-L by the appropriate averaging, =**AVERAGE(B\$2:B\$11)** for early Continental flights, =**AVERAGE(B\$12:B\$21)** for early American flights, etc.,

=AVERAGE(\$B\$2:\$C\$11) for all Continental flights,
=AVERAGE(\$B\$2:\$B\$21) for all early flights,
=AVERAGE(\$B\$2:\$C\$21) for all the flights in the sample.

- Squared deviations are then computed in columns Q-AA. These are components of sums of squares SSA, SSB, SSAB, and SSE. They are computed as

=(H2-N2)^2 in cell Q2 for SSA

=(K2-N2)^2 in cell T2 for SSB

=(E2-H2-K2+N2)^2 in cell W2 for SSAB (SS interactions)

=(B2-E2)^2 in cell Z2 for SSE (SS error, or Within SS)

- Summing these components, we get SSA, SSB, SSAB, and SSE in line 23. For example, SSA is computed in R23 as

=SUM(Q2:R21)

- Mean Squares are computed in line 25 dividing SS by df.
- F-statistics are computed in line 27 dividing the corresponding MS by MSE. For example, to test significance of the row factor (airline), we use

=R25/\$AA25

- Finally, the P-values are computed in line 28 by the command =FDIST. For example, to test significance of the airline, the P-value is computed in R28 as

=FDIST(R27,R24,\$AA24)